

```
In [121... import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [122... df = pd.read_csv('insurance.csv')
df.head()
df.head()
```

```
Out[122... 
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

```
In [123... df.shape
```

```
Out[123... (1338, 7)
```

```
In [124... df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   age         1338 non-null   int64
1   sex         1338 non-null   object
2   bmi         1338 non-null   float64
3   children    1338 non-null   int64
4   smoker      1338 non-null   object
5   region      1338 non-null   object
6   charges     1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

```
In [125... df.describe()
```

Out[125...

	age	bmi	children	charges
count	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	30.663397	1.094918	13270.422265
std	14.049960	6.098187	1.205493	12110.011237
min	18.000000	15.960000	0.000000	1121.873900
25%	27.000000	26.296250	0.000000	4740.287150
50%	39.000000	30.400000	1.000000	9382.033000
75%	51.000000	34.693750	2.000000	16639.912515
max	64.000000	53.130000	5.000000	63770.428010

In [126...

`df.columns`

Out[126...

Index(['age', 'sex', 'bmi', 'children', 'smoker', 'region', 'charges'], dtype='object')

In [127...

`df.duplicated()`

Out[127...

```

0      False
1      False
2      False
3      False
4      False
...
1333   False
1334   False
1335   False
1336   False
1337   False
Length: 1338, dtype: bool

```

In [128...

`df.isnull().sum()`

Out[128...

```

age      0
sex      0
bmi      0
children 0
smoker   0
region   0
charges  0
dtype: int64

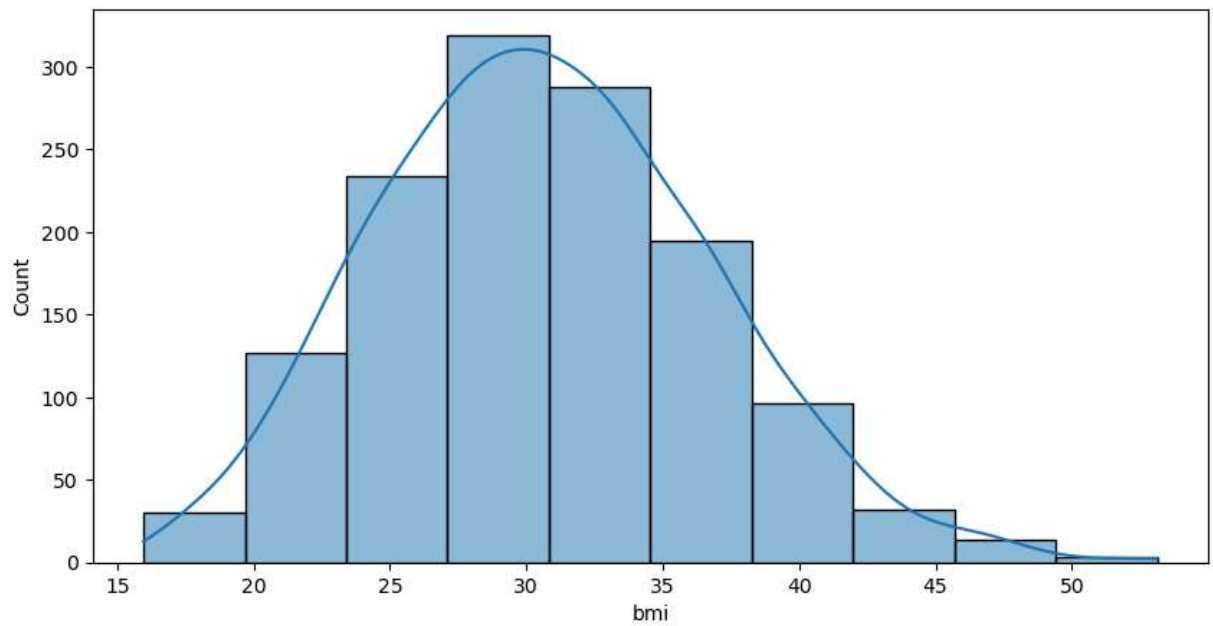
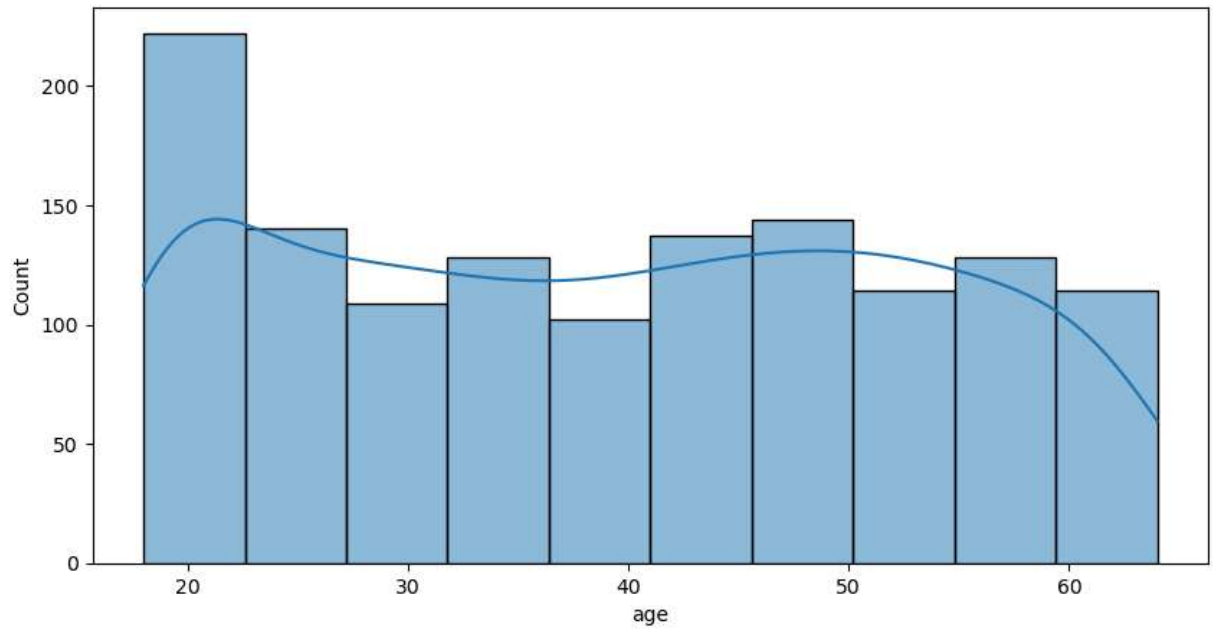
```

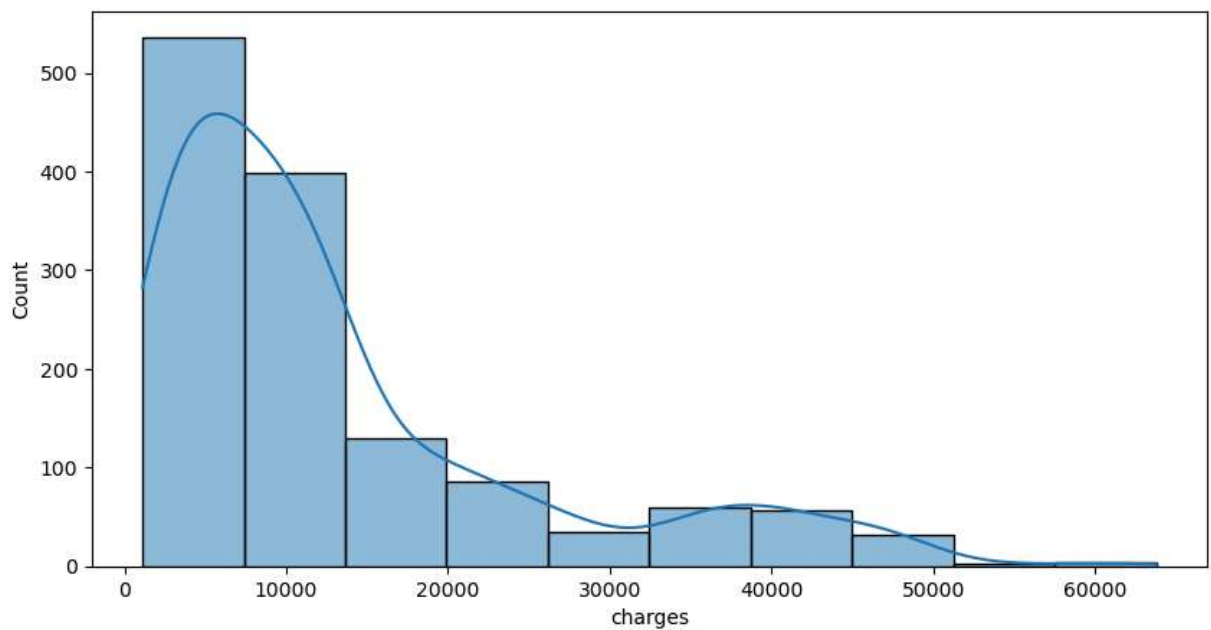
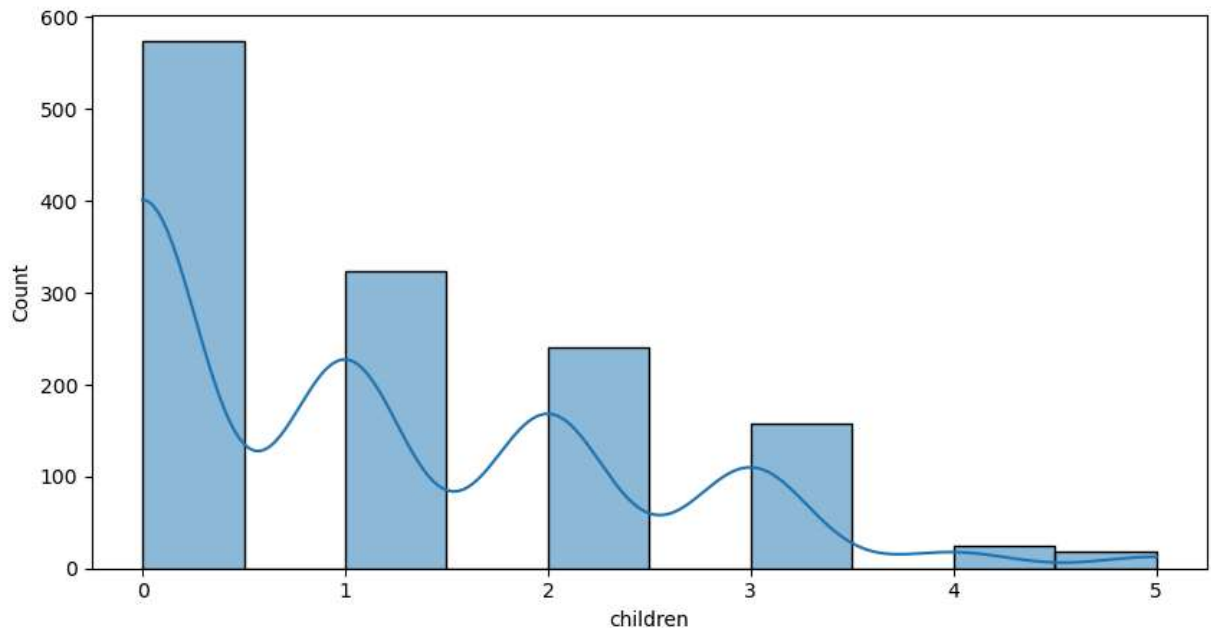
In [129...

```

numeric_col = ['age' , 'bmi' , 'children' , 'charges']
for col in numeric_col:
    plt.figure(figsize = (10 , 5))
    sns.histplot(df[col] , kde = True , bins = 10 )
    plt.show()

```





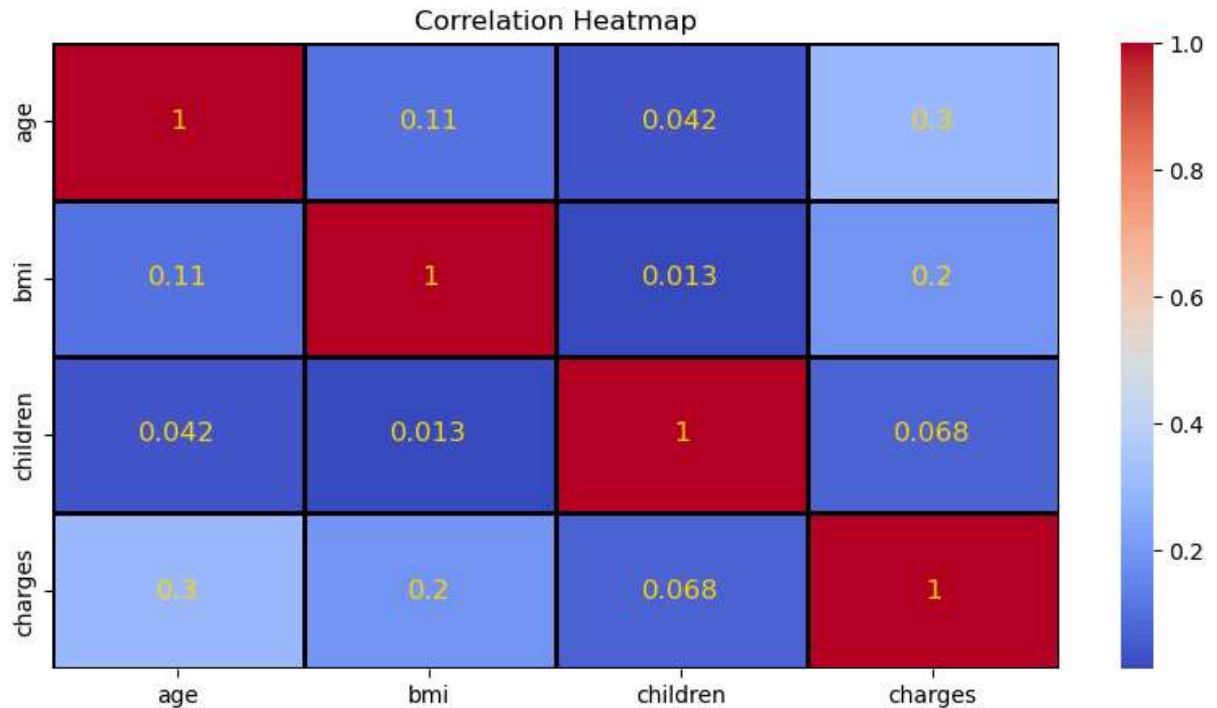
In [130...

```
#heatmap
plt.figure(figsize=(10, 5))

v = sns.heatmap(
    df[numeric_col].corr(),
    annot=True,
    cmap='coolwarm',
    annot_kws={'fontsize': 12, 'color': 'gold'},
    linewidths=2,
    linecolor='black',
    xticklabels=True,
    yticklabels=True
)

v.set(title='Correlation Heatmap')
```

```
plt.show()
```



```
In [131...] df.head()
```

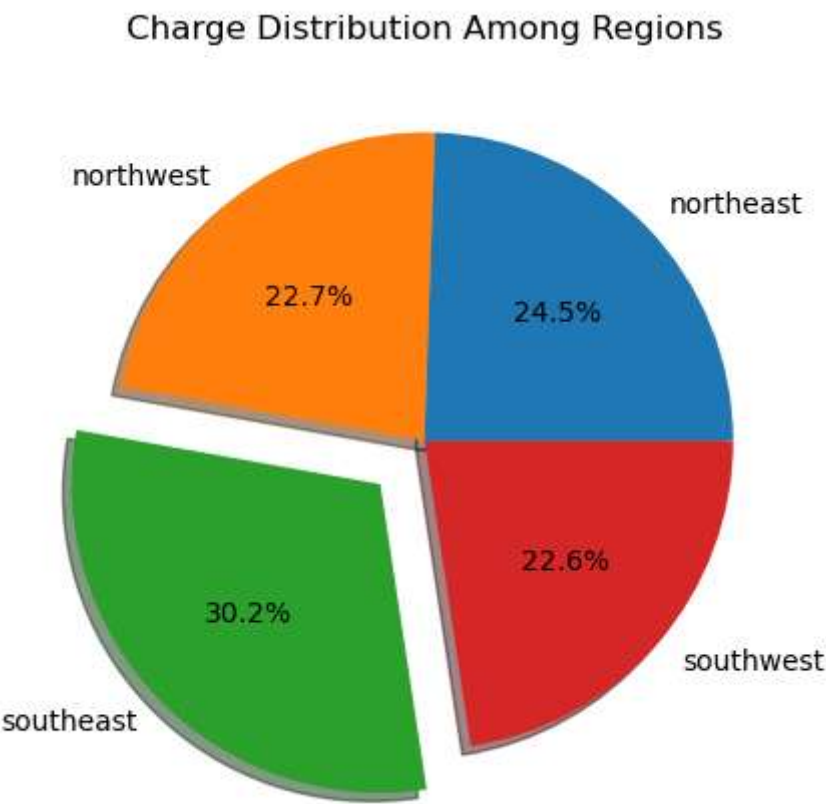
```
Out[131...]
   age  sex  bmi  children  smoker  region  charges
0   19  female  27.900      0    yes southwest  16884.92400
1   18   male  33.770      1     no  southeast  1725.55230
2   28   male  33.000      3     no  southeast  4449.46200
3   33   male  22.705      0     no  northwest  21984.47061
4   32   male  28.880      0     no  northwest  3866.85520
```

```
In [132...] df['region'].value_counts()
```

```
Out[132...]
region
southeast    364
southwest    325
northwest    325
northeast    324
Name: count, dtype: int64
```

```
In [133...]
#pie chart
plt.figure(figsize=(10, 5))
region_data = df.groupby('region')['charges'].sum()
plt.pie(
    x = region_data,
    labels=region_data.index,
    autopct='%.1f%',
    shadow=True,
```

```
explode=[0, 0, 0.2, 0]
)
plt.title('Charge Distribution Among Regions')
plt.show()
```



```
In [134... df['category_bmi'] = pd.cut(
    df['bmi'],
    bins= [0 , 18.5 , 25 ,30 , 100],
    labels=['Underweight', 'Normal', 'Overweight', 'Obese']
)
```

```
In [135... df.head()
```

Out[135...

	age	sex	bmi	children	smoker	region	charges	category_bmi
0	19	female	27.900	0	yes	southwest	16884.92400	Overweight
1	18	male	33.770	1	no	southeast	1725.55230	Obese
2	28	male	33.000	3	no	southeast	4449.46200	Obese
3	33	male	22.705	0	no	northwest	21984.47061	Normal
4	32	male	28.880	0	no	northwest	3866.85520	Overweight

```
In [136... #barchart
plt.figure(figsize=(10, 5))

sns.histplot(
    x= df['category_bmi'],
```

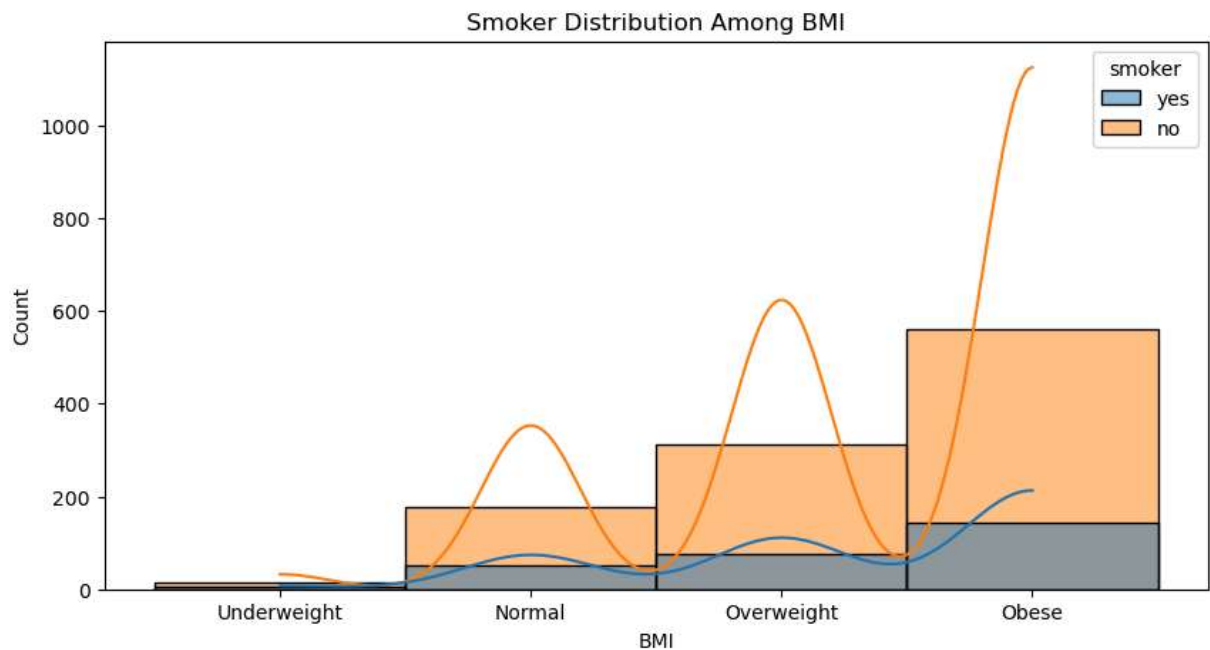
```

    hue='smoker',
    kde = True,
    bins = 10,
    data=df
)

plt.title('Smoker Distribution Among BMI')
plt.xlabel('BMI')
plt.ylabel('Count')

plt.show()

```



In [137...

```

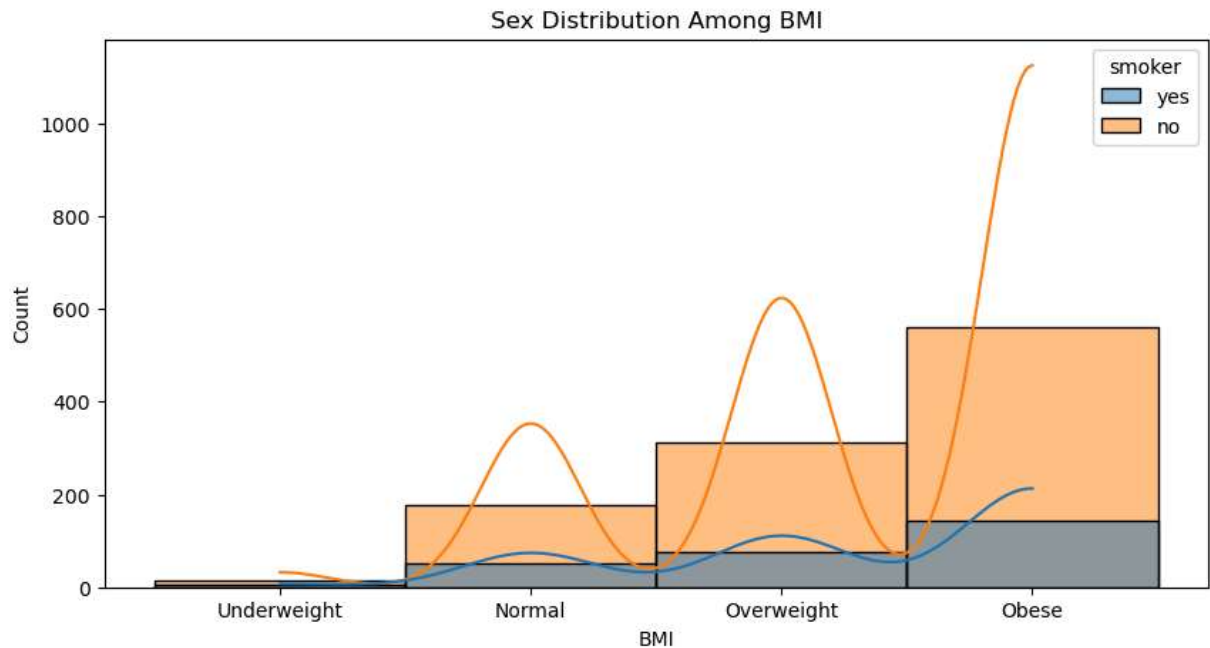
#barchart
plt.figure(figsize=(10, 5))

sns.histplot(
    x= df['category_bmi'],
    hue='smoker',
    kde = True,
    bins = 10,
    data=df
)

plt.title('Sex Distribution Among BMI')
plt.xlabel('BMI')
plt.ylabel('Count')

plt.show()

```



In [138... `df.head()`

Out[138...

	age	sex	bmi	children	smoker	region	charges	category_bmi
0	19	female	27.900	0	yes	southwest	16884.92400	Overweight
1	18	male	33.770	1	no	southeast	1725.55230	Obese
2	28	male	33.000	3	no	southeast	4449.46200	Obese
3	33	male	22.705	0	no	northwest	21984.47061	Normal
4	32	male	28.880	0	no	northwest	3866.85520	Overweight

In [139... `df['region'].isnull().sum()`

Out[139... `np.int64(0)`

In [140... `df.dtypes`

Out[140...

```
age          int64
sex          object
bmi          float64
children     int64
smoker       object
region       object
charges      float64
category_bmi category
dtype: object
```

In [141...

```
#Label encodding
# LB-> sex and smoker
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df['sex'] = le.fit_transform(df['sex'])
```



```
df['smoker'] = le.fit_transform(df['smoker'])

#on hot encoding
#OHE-> region , category_bmi
df = pd.get_dummies(df, columns=['region', 'category_bmi'])
```

In [142... df.head()

Out[142...

	age	sex	bmi	children	smoker	charges	region_northeast	region_northwest	region_south
0	19	0	27.900	0	1	16884.92400	False	False	False
1	18	1	33.770	1	0	1725.55230	False	False	False
2	28	1	33.000	3	0	4449.46200	False	False	False
3	33	1	22.705	0	0	21984.47061	False	True	False
4	32	1	28.880	0	0	3866.85520	False	True	False

In [143...

```
## Training and Testing DATA
from sklearn.model_selection import train_test_split
x = df.drop(['charges'], axis= 1)
y = df['charges']

x_train , x_test , y_train , y_test = train_test_split(x , y , test_size= 0.2 , ran
```

In [146...

```
## scaling By standardScale
from sklearn.preprocessing import StandardScaler
scale = StandardScaler()
x_train = scale.fit_transform(x_train)
x_test = scale.transform(x_test)
```

In [150... df.head()

Out[150...

	age	sex	bmi	children	smoker	charges	region_northeast	region_northwest	region_south
0	19	0	27.900	0	1	16884.92400	False	False	False
1	18	1	33.770	1	0	1725.55230	False	False	False
2	28	1	33.000	3	0	4449.46200	False	False	False
3	33	1	22.705	0	0	21984.47061	False	True	False
4	32	1	28.880	0	0	3866.85520	False	True	False

In [151... df.dtypes

```
Out[151...  age                int64
            sex                int64
            bmi              float64
            children         int64
            smoker           int64
            charges          float64
            region_northeast bool
            region_northwest bool
            region_southeast bool
            region_southwest bool
            category_bmi_Underweight bool
            category_bmi_Normal      bool
            category_bmi_Overweight  bool
            category_bmi_Obese       bool
            dtype: object
```

```
In [160... #Linear Regression
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(x_train , y_train)

L_pred = model.predict(x_test)
print('Linear Predication : ' , L_pred[:5].round(2))
```

Linear Predication : [8612.33 8818.93 35986.81 8857.04 28249.88]

```
In [164... # score of Linear
from sklearn.metrics import mean_absolute_error , mean_squared_error , r2_score
mae = mean_absolute_error(y_test , L_pred)
print("MAE : " , mae)

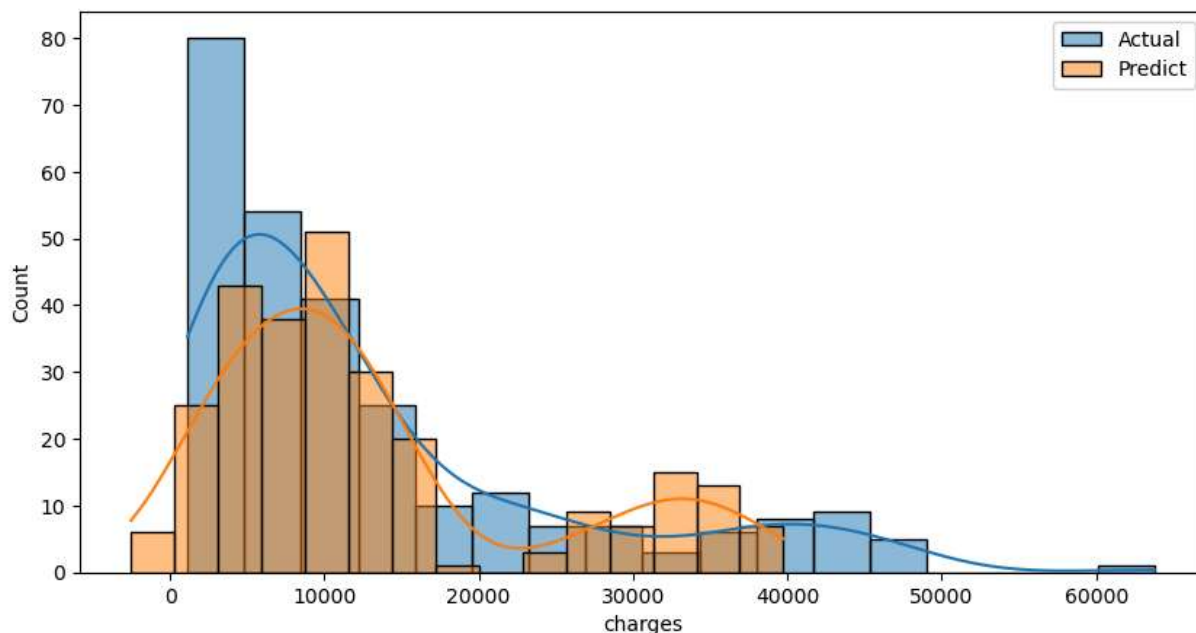
mse = mean_squared_error(y_test , L_pred)
print("MSE : " , mse)

rmse = np.sqrt(mse)
print("RMSE : " , rmse)

r2 = r2_score(y_test , L_pred)
print("R2-Score : " ,r2 )
```

MAE : 4311.331174027875
MSE : 33933847.47051578
RMSE : 5825.2766003440365
R2-Score : 0.781422707004122

```
In [171... #graph of LR
plt.figure(figsize = (10 , 5))
sns.histplot(y_test , kde = True , label = 'Actual')
sns.histplot(L_pred , kde = True , label = 'Predict')
plt.legend()
plt.show()
```



```
In [165... #Random forest Regression
from sklearn.ensemble import RandomForestRegressor

model = RandomForestRegressor()
model.fit(x_train , y_train)

R_pred = model.predict(x_test)
print('Linear Predication : ' , R_pred[:5].round(2))
```

Linear Predication : [11656.09 5175.26 28234.32 11091.92 34868.69]

```
In [168... # Scores of Forest
from sklearn.metrics import mean_absolute_error , mean_squared_error , r2_score
mae = mean_absolute_error(y_test , R_pred)
print("MAE : " , mae)

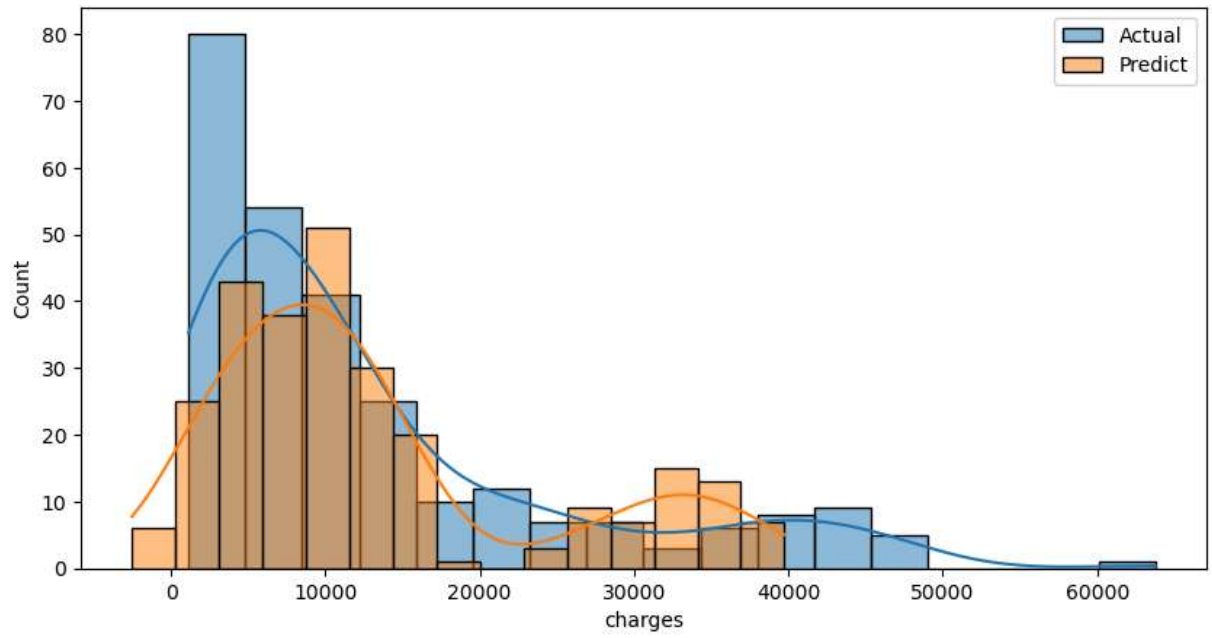
mse = mean_squared_error(y_test , R_pred)
print("MSE : " , mse)

rmse = np.sqrt(mse)
print("RMSE : " , rmse)

r2 = r2_score(y_test , R_pred)
print("R2-Score : " , r2 )
```

MAE : 2536.223695264086
MSE : 21371306.238270566
RMSE : 4622.911013449271
R2-Score : 0.8623415081533017

```
In [172... #graph of RFR
plt.figure(figsize = (10 , 5))
sns.histplot(y_test , kde = True , label = 'Actual')
sns.histplot(L_pred , kde = True , label = 'Predict')
plt.legend()
plt.show()
```



In []: